

The Brent-Price Causal Effect of the 2026 Hormuz Disruption

A Synthetic Control Approach — Milestone 3

Cherryl Chico Nikoloz Darsalia Elvis Casco

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Where we left off

Major feedbacks:

- 1 **Stronger justification that the selected donors satisfy SUTVA** — i.e. that no donor is itself contaminated by the treatment.
- 2 **Economic interpretation of why the models behaved as they did** during the treatment period
- 3 **Explore a naive counterfactual** — show what the SCM machinery buys us over simpler baselines.

Re suggested reading **Ham & Miratrix**: it concerns pre-trend identification for *matching / difference-in-differences* designs, not synthetic control on a price series, so its specific machinery does not transfer to our setting

Comment 1 — how we defend donor selection (SUTVA)

The selection question, asked of every candidate:

Does the asset co-move with Brent through common global factors, yet stay off the oil-supply / disruption path?

Approaches:

- ⇒ **(Main) Economic audit.** (Abadie 2021) Screen every candidate against two contamination channels: the **oil-supply path** (energy complex, petrocurrencies — categorically out) and the **disruption itself** (Hormuz routing, Russia/Ukraine trade — screened donor-by-donor).
- + **Statistical cross-check.** Two tests of whether a donor's *own* series shifts at T_0 — a **permutation mean-shift** test (Chow 1960) and the **HLT** trend-break test (Harvey, Leybourne & Taylor 2009).

Economic audit — Metals & Agriculturals

Metals — Silver, Platinum, Gold

- **Include:** real-rate / safe-haven / dollar channel — a monetary co-movement with Brent, off the oil-supply path.
- **Exclude:** Copper, IronOre (Russia metal supply ~4–5% world), Palladium (Russia ~40% world supply, spiked Mar-2022).

Agriculturals — Coffee, Sugar, LiveCattle

- **Include:** global trade / demand and the dollar; producers outside the conflict zones (Brazil, India, US).
- **Exclude:** Wheat, Corn (Russia+Ukraine ~30% / 13% world exports), Soybeans (grain cross-elasticity), Cotton (oil→polyester substitution).

Economic audit — Equities & FX

Equities — SP500, Nikkei

- **Include:** global growth and the equity risk premium — the demand channel shared with oil; no Russia or Gulf-energy weight.
- **Exclude:** EM_Eq (Russia index weight); energy-heavy DAX / FTSE / TSX-60 never enter.

FX — AUD, JPY, CHF, CNY, INR, KRW, ZAR, MXN

- **Include:** safe-haven (JPY, CHF), EM-growth (ZAR, MXN, INR, KRW), China-demand (CNY, AUD) — dollar and growth factors.
- **Exclude:** EUR, DXY (EU energy-crisis / dollar on the invasion), GBP (UK gas); petrocurrencies (CAD, NOK, BRL, ...) never enter.

Economic audit — Rates / credit & Volatility

Rates / credit — TLT, HYG

- **Include:** TLT (US duration) and HYG (credit-risk premium) — the discount-rate / risk side of the global factor, not oil-linked.
- **Exclude:** US10Y (Russia: inflation-expectations / flight-to-quality loads it onto the treatment).

Volatility — VIX

- **Include:** the single risk-on / risk-off switch — the fear premium that also lifts oil in stress.
- **Exclude:** none — one volatility proxy, kept deliberately to avoid over-weighting the risk channel.

Net of the audit: 19 of 32 candidates retained (shared pool, both events — 13 dropped for Russia exposure).

Statistical cross-check — shifts at T_0

- **Permutation mean-shift** (Chow 1960): split the donor's own series at T_0 , take the mean gap (after – before), and compare it to thousands of random before/after relabelings — an unusually large gap signals a real shift. Nonparametric, no distributional assumption.
- **HLT trend-break** (Harvey, Leybourne & Taylor 2009): tests whether the donor's own *trend* breaks at T_0 , valid whether the price is stationary or a random walk. Brent-free, so a break implicates the donor itself, not Brent's own T_0 move.

Flagged at T_0 (of 19 retained)	Hormuz 2026	Russia 2022
HLT trend-break	0 / 19	0 / 19
Permutation mean-shift	0 / 19	1 / 19 (CNY)

HLT flags nothing. The lone flag — **CNY** — is the concurrent China zero-COVID reopening, not Russia-treatment, and is kept on the audit's judgment. *Tests inform; the audit decides.*

Comment 2 — why the models behaved as they did

The question:

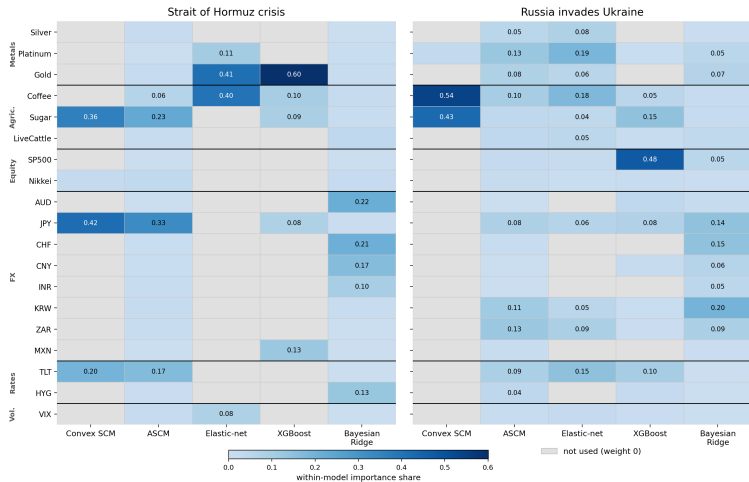
*Why did the five models move (and disagree) the way they did during the event — and does that behaviour **inflate** the premium?*

How we answer it — two steps:

- 1 **Behaviour = donor weights.** Read which donors each model leans on. This explains *why* there is a gap at all (the weighted donors track common factors, *not* the oil-specific shock) and *why* the models diverge on Russia (different donors; one degenerate model).
- 2 **Sign the contamination.** The natural worry — are those donors themselves moved by the treatment (demand destruction)? We split each gap into a clean part and a donor-contamination part and *sign* it: conservative for Hormuz (a **lower bound**), positive for Russia (why Russia is only a **magnitude check**).

Take-away: the behaviour is interpretable (donor-driven), and the residual contamination is signed — it does not inflate the headline.

Comment 2 — how the donor weights drive model behaviour



Hormuz. Weight on safe-haven / non-oil donors (Gold, JPY, Sugar) — they track common risk but *not* the chokepoint shock $\Rightarrow$ synthetic undershoots Brent $\Rightarrow$ positive gap.

Russia — divergence is donor-driven. Models lean on *different* donors (Coffee/Platinum vs SP500); Bayesian Ridge degenerates ($w \approx 1.8-2.4$) $\Rightarrow$ dropped from the IQR.

Each column = that model's own importance metric (convex weight / $|\hat{\beta}_j|$ / XGBoost gain), normalised over the 19 donors. Gray = not used.

Comment 2 — the contamination bias is signed, and it makes us conservative

The critique. If high oil *crowds out demand*, activity-proxy donors are pushed *down*, dragging the synthetic down → contamination in the gap. **Which way, and how big?** We sign each linear model's net donor-bias ($-\sum_j w_j \delta_j$) over the full post-window:

Model	Hormuz net bias	Russia net bias
Convex SCM	−1.3% (conservative)	+21.4%
ASCM	−2.1% (conservative)	+12.7%
Elastic-net	−2.4% (conservative)	+20.6%
Bayesian Ridge	−7.6% (conservative)	−7.1%

Linear models only (XGBoost nonlinear). Source: `contamination_bias_decomposition.csv`.

- **Hormuz: bias negative** for all linear models → demand-destruction makes the SCM *under*-state the premium, so the headline **~59%** is a **lower bound**.
- **Russia: bias positive** → retained cyclical donors (Coffee, Platinum, SP500) were pushed *down*, dragging the synthetic down and *inflating* the gap → Russia is a magnitude check, not the headline.

Statistical test 1 — in-space placebo

Procedure (Abadie 2010): refit the SCM treating each donor as treated; rank Brent's post/pre RMSPE ratio against the $N=20$ units (Brent + 19 donors). **Permutation** $p = \text{rank} / N$; floor $p = 1/20 = 0.05 \rightarrow$ rank 1 is the strongest signal the 19-donor pool can produce.

Model	Hormuz 2026	Russia 2022
	rank $\rightarrow p$	rank $\rightarrow p$
Convex SCM	1 $\rightarrow$ 0.05	10 $\rightarrow$ 0.50
ASCM	1 $\rightarrow$ 0.05	12 $\rightarrow$ 0.60
Elastic-net	1 $\rightarrow$ 0.05	8 $\rightarrow$ 0.40
XGBoost	1 $\rightarrow$ 0.05	4 $\rightarrow$ 0.20
Bayesian Ridge	1 $\rightarrow$ 0.05	12 $\rightarrow$ 0.60

Source: `data/validation/inference_inspace_brent_summary.csv` (19-donor shared pool).

- **Hormuz:** Brent ranks 1 for **all five models** $\rightarrow$ its break beats every placebo; strongest signal the pool allows.
- **Russia:** mid-pack (rank 8–12) for the linear models; XGBoost is best at rank 4 ($p=0.20$) but still not significant — consistent with Russia as a magnitude check, not a clean inferential case.

Statistical test 2 — leave-one-out (LOO)

Procedure. Drop one influential donor ($w > 5\%$) at a time, refit, recompute the ATT. Report the **range** (max — min, pp) across the leave-one-donor-out refits — a small range means no single donor drives the result.

Model	Hormuz LOO range (pp)	Russia LOO range (pp)
Convex SCM	8.2	26.8
ASCM	5.7	19.1
Elastic-net	6.7	18.8
XGBoost	6.1	13.8
Bayesian Ridge	10.4	27.3 [†]

Source: `data/validation/inference_loo_{event}_{model}.csv`.

- **Hormuz: tight** (~6–10 pp on a ~59% effect) — no single donor flips the result; ASCM tightest.
- **Russia: wide** (14–27 pp). Bayesian Ridge ([†] range crosses zero, −21.9 to +5.4) is LOO-fragile, consistent with its degenerate fit; XGBoost is the most stable. Reinforces Russia as a magnitude check, not the headline.

Comment 3 — naive counterfactual baselines

Question. What does the donor-informed SCM buy us over baselines that ignore the donor pool entirely? We compare the ensemble-median SCM premium against four naive counterfactuals at the full post-window.

Counterfactual	Hormuz %	Russia %
SCM ensemble (donor-informed)	58.9	22.0
Random walk — flat	51.9	8.7
Random walk — drift	52.6	−6.9
Linear trend	73.8	−4.2
Interrupted time series (ITS)	73.8	−4.2

Source: `data/validation/naive_baseline_comparison.csv` (full-window horizon).

Reading.

- **Random walks understate** the premium — they freeze the counterfactual at the last pre-event level and miss the donor-implied path the rest of the market was on.
- **Linear trend / ITS overstate** Hormuz (74%) and *flip sign* on Russia (−4%) — naive extrapolation of a pre-trend is spurious when the donor factor itself turned.
- The SCM sits **between** the two failure modes: anchored to a *data-driven* counterfactual that tracks the global factor without freezing or blindly extrapolating it.

Summary — headline ATT across post-event windows

- 1 **SUTVA** — audit (oil-supply / disruption path), corroborated by the breakpoint test.
- 2 **Model behaviour** — weight on risk donors explains the gap; contamination bias signed (Hormuz conservative).
- 3 **Naive baselines** — random walks understate, trend / ITS over-extrapolate; the SCM sits between.

PLACEHOLDER — postwindow_sensitivity.png
(regenerate via notebooks/08_Milestone3_Plots.ipynb)

Thank you

Questions and discussion

github.com/ElvisCasco/Brent_Analysis